

SPRING DESIGN CALCULATION

The spring parameters are selected as follows:

$$L_o = 100 \text{ mm}$$

$$\delta = 60 \text{ mm}$$

$$D_o = 20 \text{ mm}$$

$$d = 1.5 \text{ mm}$$

$$D = D_o - d$$

$$= 20 - 1.5$$

$$= 18.5 \text{ mm}$$

The spring index is:

$$C = D/d = 12.3$$

Wahl correction factor:

$$K = ((4C-1)/(4C-4)) + (0.615/C) = 1.1164$$

Shear stress equation:

$$\tau = (K8PC)/(\pi d^2)$$

$$450 = (1.1164 \times 8 \times P \times 12.3)/(\pi \times 1.52)$$

$$P = 29 \text{ N}$$

For allowable stress $\tau = 450 \text{ MPa}$:

$$P = 29 \text{ N}$$

Spring stiffness:

$$P/\delta=29/60=0.4825 \text{ N/mm}$$

Theoretical stiffness:

$$k_{act}=Gd^4/8D^3N$$

$$=(81370 \times 1.5^4)/(8 \times 18.5^3 \times 20)$$

$$=0.4066 \text{ N/mm}$$